CateDocs: A New Visual Analytics Approach to Exploring High Dimensional Categorical Datasets

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Introduction

Categorical dataset Challenge: Categorical datasets with large numbers of dimensions and categories are common in real applications. Many tasks need to be addressed, such as effectively guiding users to navigate in the high dimensional space and supporting cluster analysis, prediction, and other investigative tasks. However, most existing visualization approaches do not scale well to large, high dimensional categorical datasets.

CateDocs: Our strategy is to regard category values as a vocabulary and then each data item becomes a document built upon this vocabulary. This allows a categorical dataset to be studied as a corpus. Advanced document analysis and visualization techniques can then be used for visual analytics of categorical datasets. They excel in handling high dimensional keyword spaces, processing large numbers of documents, dealing with missing values, and presenting semantics, all of which are desired in high dimensional categorical data analysis.

Topic Modeling Analysis: We borrow Latent Dirichlet Allocation (LDA) [1] for subspace cluster analysis. LDA automatically identifies topics from the document corpus converted from a categorical dataset. The keywords defining a topic are then mapped to a subspace cluster and the documents closely related to the topic are mapped to member data items of the cluster.

Advantages of the approach include:

- LDA is able to reveal topics widely distributed in a corpus as well as salient topics in relatively small subsets of documents. This allows user to identify small but significant clusters;
- LDA scales well to the number of dimensions and categories as well as the record size;
- The resulting clusters / subspaces are described by a set of category values, which are very intuitive to interpret and visualize.

Risk/preventive Rule Analysis: Risk rule mining helps analysts understand the relationship between a dependent variable and many independent variables. However, applying it directly to large dataset is costly.

We propose:

1.Using LDA to find meaningful lower dimensional subspaces for risk/preventive rule mining.

2.Using Optimal Risk Rule Mining [4] to discover rules from the reduced subspaces constructed according to LDA results.

Results of our experiment (Figure 2) show that very few rules were missing while the time cost was reduced significantly using this approach.

Convert a Categorical Dataset to a Document Corpus

Each record in a dataset is converted to a "document" represented by a bag of category values. For example:

User Information			Movie Inmation		
Age (7)	Gender (2)	Occupation (21)	Title (3883)	Year (180)	Genres (40)
18-24	M	self-employed	The Matrix	90s	Action Sci-Fi Thriller
25-34	M		Blast from the Past	90s	Comedy Romance
25-34	M	artist	Malice	90s	Thriller
⋮	⋮	⋮	⋮	⋮	⋮
18-24	M		Batman & Robin	90s	Action Adventure Crime

Doc #1:Age=18-24 gender=M occupation=self-employed title=The_Matrix year=90s genres=Action genres=Sci-Fi genres=Thriller.

Doc #2: Age=25-34 gender=M title=Blast_from_the_Past year=90s genres=Comedy genres=Romance

Doc #3: Age=25-34 gender=M occupation=artist title=Malice year=90s genres=Thriller

…

Doc #N: Age=18-24 gender=M title=Batman&Robin year=90s genres=Action genres=Adventure genres=Crime

In the example above, the movie genres dimension may have multiple values. Each value becomes an individual keyword in the document. The user occupation may miss so no keyword for this dimension is included in the corresponding document.

Cluster Visual Analytics

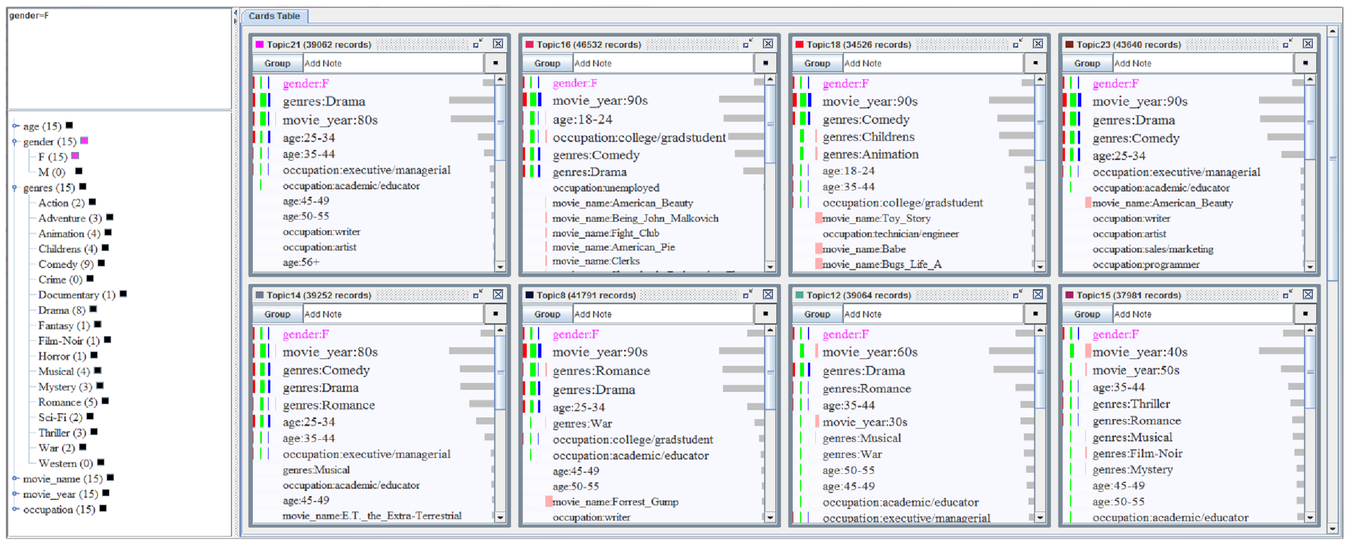


Figure 1: Clusters containing category value gender=F in the MovieLens dataset. The category values in the clusters demonstrate the subtle difference among the clusters.

Identify clusters:

1. Create a corpus from a categorical dataset.
2. Create a LDA topic model, in the example we use Stanford Topic Modeling Toolbox (STMT) [5].
3. Map each topic to a cluster. The cluster is defined by the top 20 category values most closely related to the topic. Since high frequency category values can gain high weights in many topics, we remove a category value from a topic if its local support (the ratio of data items with the keyword in the cluster) is lower than its global support (the ratio of data items with the category value in the dataset). Data items with higher weights than a threshold on a topic are considered members of the corresponding cluster.

Determining a proper topic number for LDA is remained for future work. Alternatively, LDA is executed several times with different specified numbers of topics, which allows users to examine either coarse or fine clusters by simply loading the results of model runs with different numbers of topics.

Visualization:

Each Topic Card (Figure 1) represents a cluster.

- Cards are initially order by cluster interestingness, which is measured by the concentration of data item weights on the topic
- The list in a card contains category values defining the subspace cluster.
- Colorful blocks on the left of a category value display statistics of the value, such as probability of occurrence locally and globally.
- Grey bar on the right reveals the weight of the value in the topic.
- In the menu on the left, category values are organized as leaf nodes of the dimensions, enabling user to filter cluster by clicking and highlight value by coloring. It establishes a connection between the clusters and the dimensions.

Two case studies on real world high dimensional datasets have been conducted. A video demo presenting these case studies can be found from http://www.youtube.com/watch?v=W2Kt7WKIMTI

Risk/Preventive Rule Mining

We compared two approaches on the Mushroom dataset (8,214 data items, 22 independent variables, and 1 dependent variable) [2]. The first one (ORRM-only) directly applies Optimal Rule Mining to the dataset. The second one (LDA-ORRM) consists of five steps:

1. Convert data items to documents.
2. Apply LDA to the corpus with M topics specified.
3. Filter out topics in which the dependent variable has low probability.
4. Merge categorical values highly related to the remaining topics to create a subspace.
5. Project the dataset to the subspace and apply Optimal Rule Mining to it.

In the experiment, LDA-ORRM was conducted multiple times with different topic numbers. Results are reported in Figure 2. Let Set A be the number of distinct category values in the rules generated by a LDA-ORRM run. Set B is defined as the number of distinct category values in the rules generated by the ORRM-only approach. The running time of a LDA-ORRM approach is the total running time of steps 2-5. Three measures were calculated: (1) Time ratio = LDA-ORRM running time / ORRM-only running time; (2) Precision = Size of (Set A ∩ Set B) / Size of Set B; (3) Recall = Size of (Set A ∩ Set B) / Size of A. The results (Figure 2) suggests that LDA-ORRM was much faster than ORRM-only with a small amount of information loss when the topic number is larger than or equals to 20.

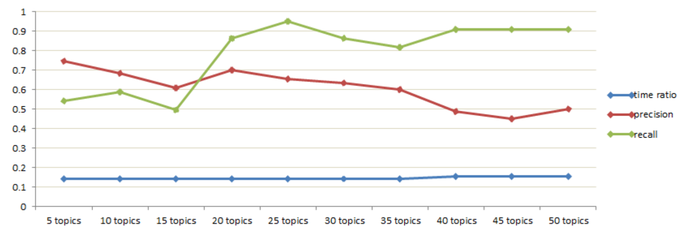


Figure 2: The results of the experiment comparing ORRM-only and LDA-ORRM approaches.

References

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